

Multi – level Linked List – Print List(Main List and Child)
–Masters and Akrabazzi failure in Time Complexity Calculation

Masters Theorem

$$2T\left(\frac{N-1}{2}\right) + c_1 \times k + c_2, \text{ Base Case: } T(0) = 1.$$

where,

k = recursion depth.

Master Theorem requires

$$T(N) = aT\left(\frac{N}{b}\right) + f(N),$$

where f(N) is a function of the current problem size N.

But our additive term is:

$$c_1 k + c_2,$$

and `k` is not the current subproblem size.

At depth k,

$$N_k = \frac{N - (2^k - 1)}{2^k}$$

Multiply by 2^k

$$N_k 2^k = N - (2^k - 1)$$

$$N_k 2^k = N - 2^k + 1$$

$$N_k 2^k + 2^k = N + 1$$

Factor:

$$2^k(N_k + 1) = N + 1$$

$$2^k = \frac{N + 1}{N_k + 1}$$

Take $\log_2$

$$\log_2 2^k = \log_2 \left(\frac{N + 1}{N_k + 1} \right)$$

$$k = \log_2 \left(\frac{N + 1}{N_k + 1} \right)$$

Using

$$\log \frac{a}{b} = \log a - \log b$$

We get:

$$k = \log_2(N + 1) - \log_2(N_k + 1)$$

Notice k depends on:

- *the original input size N ,*
- *the current subproblem size N_k .*

Thus it is not a function solely of the current subproblem size.

Lets say,

Original input , $N = 31$.

Suppose the current subproblem size is: $N_k = 7$.

Then : $k = \log_2(32) - \log_2(8) = 5 - 3 = 2$.

Cost: $2c_1 + c_2$.

Original input , $N = 127$.

Same current subproblem size is: $N_k = 7$.

Then : $k = \log_2(128) - \log_2(8) = 7 - 3 = 4$.

Cost: $4c_1 + c_2$.

Therefore there is no single function:

$f(N_k)$

that represents the additive work.

That is why:

$$2T\left(\frac{N-1}{2}\right) + c_1 \times k + c_2$$

does not fit the standard Master Theorem form.

Similarly,

Akra – Bazzi requires :

$$T(x) = \sum a_i T(b_i(x) + h_i(x)) + g(x)$$

where

$g(x)$ must be a function of the current subproblem size x .

In the equation:

$$2T\left(\frac{N-1}{2}\right) + c_1 \times k + c_2$$

At depth k ,

$$cost = c_1 k + c_2$$

Now solve for k :

$$k = \log_2(N + 1) - \log_2(N_k + 1)$$

Substitute:

$$g = c_1(\log_2(N + 1) - \log_2(N_k + 1)) + c_2$$

$$i. e. g(N, N_k) = c_1(\log_2(N + 1) - \log_2(N_k + 1)) + c_2$$

Notice the first term:

$$\log_2(N + 1)$$

This is the original problem size, not the current subproblem size.

So, the cost is not determined by N_k alone.

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Original input , $N = 31$.

Suppose the current subproblem size is: $N_k = 7$.

$$\text{Then : } k = \log_2(32) - \log_2(8) = 5 - 3 = 2.$$

$$\text{Cost: } 2c_1 + c_2 .$$

Original input , $N = 127$.

Same current subproblem size is: $N_k = 7$.

Then : $k = \log_2(128) - \log_2(8) = 7 - 3 = 4$.

Cost: $4c_1 + c_2$.

Notice what happened:

$$N_k = 7$$

in both cases.

But the costs are different:

$$2c_1 + c_2$$

vs

$$4c_1 + c_2.$$

Therefore the cost cannot be expressed as

$$g(N_k).$$

The same subproblem size produces different costs depending on where it appears in the recursion tree.

That is exactly why:

$$\boxed{g = c_1k + c_2}$$

*is not solely a function of the current subproblem size, and
therefore it does not fit the standard assumptions of
Master Theorem or Akra – Bazzi.*
